



Experimental Characterization of a Helical Resonator for Impedance Matching in a Trapped – Ion RF System

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1) Experimental Setup

The experiment utilized a SIGLENT SVA 1032X Vector Network Analyzer (VNA) to perform S11 measurements of a helical resonator. The VNA was configured to sweep the frequency range from 24.3 MHz to 25.1 MHz. This setup allowed for the characterization of the resonator's impedance matching properties.

2) Measurements and Results

The helical resonator exhibited a resonance frequency at approximately 24.73 MHz. The best loss achieved was -29.58 dB, occurring precisely at 24.73000 MHz, indicating a strong impedance match at this frequency. The 3 dB bandwidth of the resonance was measured to be 92 kHz, with band edges observed at 24.684 MHz and 24.776 MHz.

Further analysis of the Smith chart revealed the impedance at resonance to be

M1: $48.99\ \Omega$, $-747.42\ \text{m}\Omega$, $8.61\ \text{Nf}$ at $24.732\ \text{MHz}$.

Additional markers on the Smith chart provided insights into the impedance at other frequencies:

M2:

$24.688584\ \text{MHz}$, $9.23\ \Omega$, $11.30\ \Omega$, $72.87\ \text{nH}$

M3:

$24.775416\ \text{MHz}$, $10.90\ \Omega$, $-25.08\ \Omega$, $256.15\ \text{pF}$

The loaded **Q factor** for the resonator was calculated to be **268.80**

3) Discussion

The measurements demonstrate that the helical resonator is well-matched to a nominal $50\ \text{ohm}$ system at its resonance frequency. The achieved return loss of $-29.52\ \text{dB}$ signifies minimal signal reflection, which is critical for efficient power transfer. The loaded Q factor of 268.80 suggests a relatively sharp resonance, allowing for precise tuning and stable operation. The impedance at resonance, being close to $50\ \text{ohms}$ with a small reactive component, confirms that this resonator is suitable for delivering stable radiofrequency (RF) power to an ion-trap setup.

4) Conclusion

The helical resonator exhibits excellent impedance matching characteristics near its resonance frequency of $24.73\ \text{MHz}$. The low return loss and proximity of the resonant impedance to $50\ \text{ohms}$ indicate that it is a suitable component for RF delivery in an ion-trap system, ensuring efficient and stable operation.

5) Measurement Images

IMG_2318.jpeg: [Smith chart and impedance markers \(Image file reference: IMG_2318.jpeg\)](#)

IMG_2320.jpeg: [Return loss, bandwidth, and Q factor \(Image file reference: IMG_2320.jpeg\)](#)



